

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

```
In [2]: df = pd.read_csv("netflix_titles.csv")

df.head()
```

```
Out[2]:
```

	show_id	type	title	director	cast	country	date_added	release_year
--	---------	------	-------	----------	------	---------	------------	--------------

0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
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1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
---	----	---------	---------------	-----	---	--------------	--------------------	------

2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
---	----	---------	-----------	-----------------	---	-----	--------------------	------

3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
---	----	---------	-----------------------	-----	-----	-----	--------------------	------

4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
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```
In [3]: print("Shape:", df.shape)

df.info()

df.describe()
```

```
Shape: (8807, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8807 non-null   object
 1   type            8807 non-null   object
 2   title           8807 non-null   object
 3   director        6173 non-null   object
 4   cast            7982 non-null   object
 5   country         7976 non-null   object
 6   date_added      8797 non-null   object
 7   release_year    8807 non-null   int64
 8   rating          8803 non-null   object
 9   duration        8804 non-null   object
10   listed_in       8807 non-null   object
11   description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Out[3]: **release_year**

count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

In [4]: `df.isnull().sum()`

```
Out[4]: show_id      0
        type        0
        title       0
        director    2634
        cast        825
        country     831
        date_added   10
        release_year 0
        rating       4
        duration     3
        listed_in    0
        description  0
        dtype: int64
```

```
In [5]: df['director'].fillna('Unknown', inplace=True)
        df['cast'].fillna('Not Available', inplace=True)
        df['country'].fillna('Unknown', inplace=True)

        df.dropna(subset=['rating'], inplace=True)
```

C:\Users\M Rushika Krishna\AppData\Local\Temp\ipykernel_22712\3602245055.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['director'].fillna('Unknown', inplace=True)
```

C:\Users\M Rushika Krishna\AppData\Local\Temp\ipykernel_22712\3602245055.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['cast'].fillna('Not Available', inplace=True)
```

C:\Users\M Rushika Krishna\AppData\Local\Temp\ipykernel_22712\3602245055.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

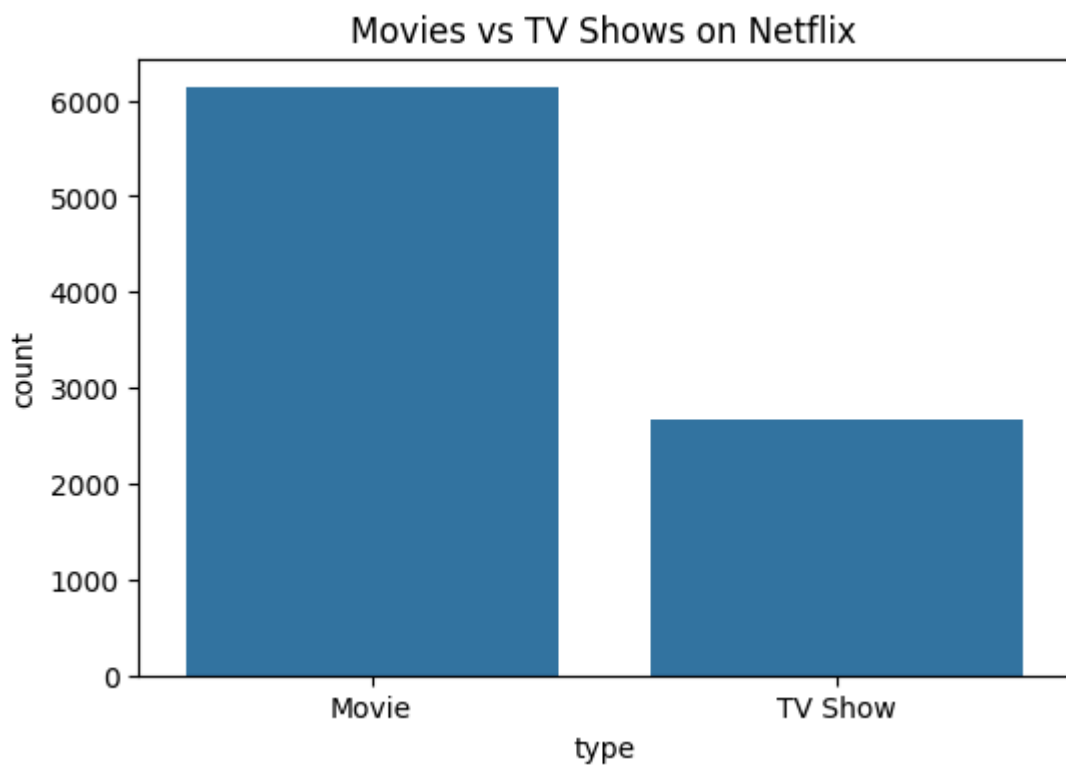
```
df['country'].fillna('Unknown', inplace=True)
```

```
In [6]: df.describe(include='all')
```

Out[6]:

	show_id	type	title	director	cast	country	date_added	release_year
count	8803	8803	8803	8803	8803	8803	8793	8803.000000
unique	8803	2	8803	4528	7690	749	1766	NaN
top	s8807	Movie	Zubaan	Unknown	Not Available	United States	January 1, 2020	NaN
freq	1	6129	1	2631	825	2818	109	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.179825
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.821247
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.000000
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2021.000000

```
In [7]: plt.figure(figsize=(6,4))
sns.countplot(x='type', data=df)
plt.title("Movies vs TV Shows on Netflix")
plt.show()
```



```
In [8]: df['date_added'] = pd.to_datetime(df['date_added'])

df['year_added'] = df['date_added'].dt.year

plt.figure(figsize=(12,5))

sns.countplot(
    x='year_added',
    data=df,
    order=df['year_added'].value_counts().index.sort_values()
)

plt.xticks(rotation=90)

plt.title("Content Added Each Year")

plt.show()
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[8], line 1
----> 1 df['date_added'] = pd.to_datetime(df[
      3 df['year_added'] = df['date_added'].dt.year
      5 plt.figure(figsize=(12,5))

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\tools\datetimes.py:1068, in to_datetime(arg, errors, dayfirst, yearfirst, utc, format, exact, unit, infer_datetime_format, origin, cache)
    1066         result = arg.tz_localize("utc")
    1067 elif isinstance(arg, ABCSeries):
-> 1068     cache_array = _maybe_cache(arg, format, cache, convert_listlike)
    1069     if not cache_array.empty:
    1070         result = arg.map(cache_array)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\tools\datetimes.py:249, in _maybe_cache(arg, format, cache, convert_listlike)
    247 unique_dates = unique(arg)
    248 if len(unique_dates) < len(arg):
-> 249     cache_dates = convert_listlike(unique_dates, format)
    250     # GH#45319
    251     try:

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\tools\datetimes.py:435, in _convert_listlike_datetimes(arg, format, name, utc, unit, errors, dayfirst, yearfirst, exact)
    433 # `format` could be inferred, or user didn't ask for mixed-format parsing.
    434 if format is not None and format != "mixed":
-> 435     return _array_strptime_with_fallback(arg, name, utc, format, exact, errors)
    437 result, tz_parsed = objects_to_datetime64(
    438     arg,
    439     dayfirst=dayfirst,
    (...) 443     allow_object=True,
    444 )
    446 if tz_parsed is not None:
    447     # We can take a shortcut since the datetime64 numpy array
    448     # is in UTC

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\tools\datetimes.py:469, in _array_strptime_with_fallback(arg, name, utc, fmt, exact, errors)
    458 def _array_strptime_with_fallback(
    459     arg,
    460     name,
    (...) 464     errors: str,
    465 ) -> Index:
    466     """
    467     Call array_strptime, with fallback behavior depending on 'errors'.
    468     """
-> 469     result, tz_out = array_strptime(arg, fmt, exact=exact, errors=errors,
    utc=utc)
    470     if tz_out is not None:
    471         unit = np.datetime_data(result.dtype)[0]

File pandas\_libs\tslibs\strptime.pyx:501, in pandas._libs.tslibs.strptime.array_strptime()

```

File pandas/_libs/tslibs/strptime.pyx:451, in pandas._libs.tslibs.strptime.array_strptime()

File pandas/_libs/tslibs/strptime.pyx:583, in pandas._libs.tslibs.strptime._parse_with_format()

ValueError: time data " August 4, 2017" doesn't match format "%B %d, %Y", at position 1441. You might want to try:

- passing `format` if your strings have a consistent format;
- passing `format='ISO8601'` if your strings are all ISO8601 but not necessarily in exactly the same format;
- passing `format='mixed'`, and the format will be inferred for each element individually. You might want to use `dayfirst` alongside this.

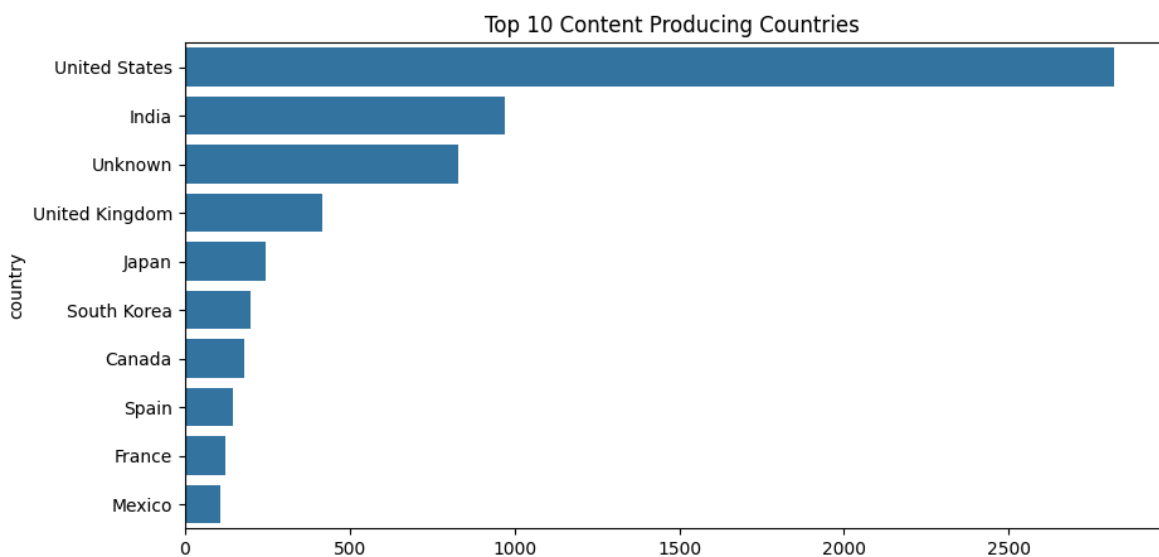
```
In [9]: top_countries = df['country'].value_counts().head(10)
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(
    x=top_countries.values,
    y=top_countries.index
)
```

```
plt.title("Top 10 Content Producing Countries")
```

```
plt.show()
```

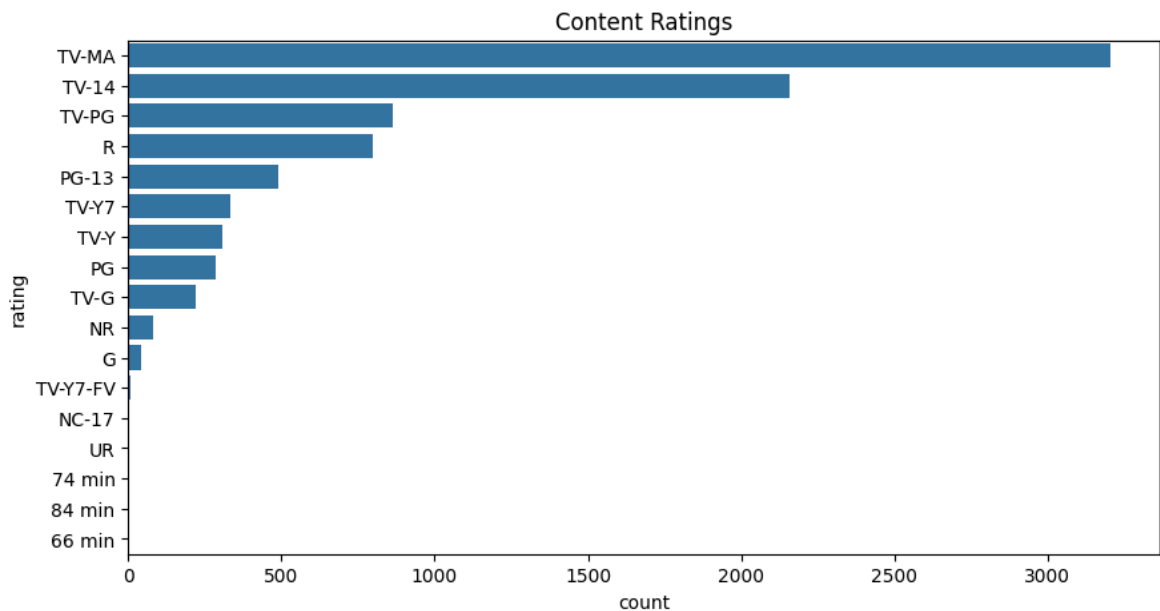


```
In [10]: plt.figure(figsize=(10,5))
```

```
sns.countplot(
    y='rating',
    data=df,
    order=df['rating'].value_counts().index
)
```

```
plt.title("Content Ratings")
```

```
plt.show()
```

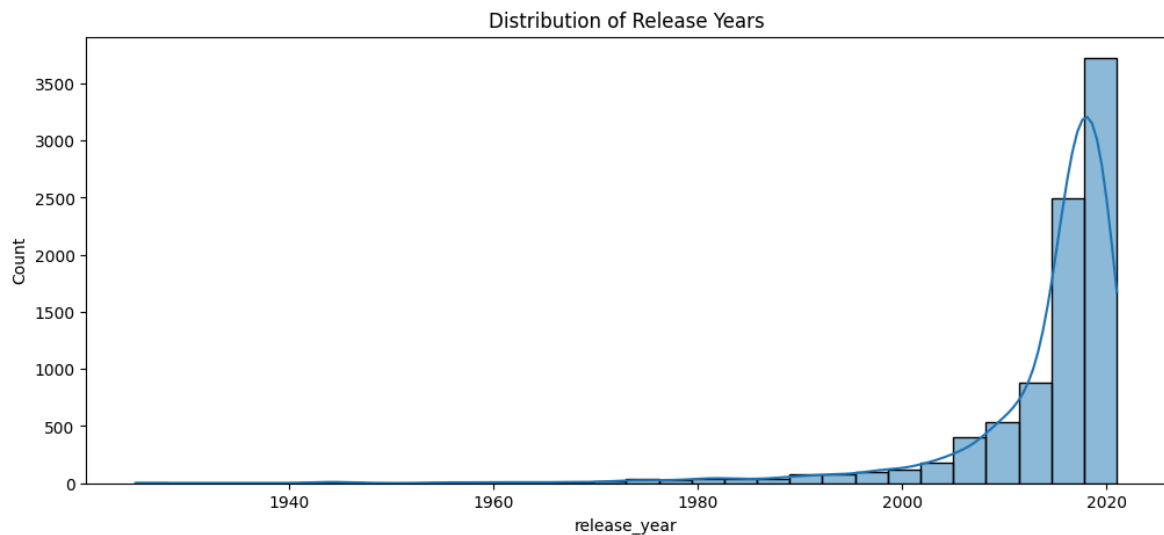


```
In [11]: plt.figure(figsize=(12,5))

sns.histplot(
    df['release_year'],
    bins=30,
    kde=True
)

plt.title("Distribution of Release Years")

plt.show()
```



```
In [12]: numeric_df = df.select_dtypes(include=['int64','float64'])

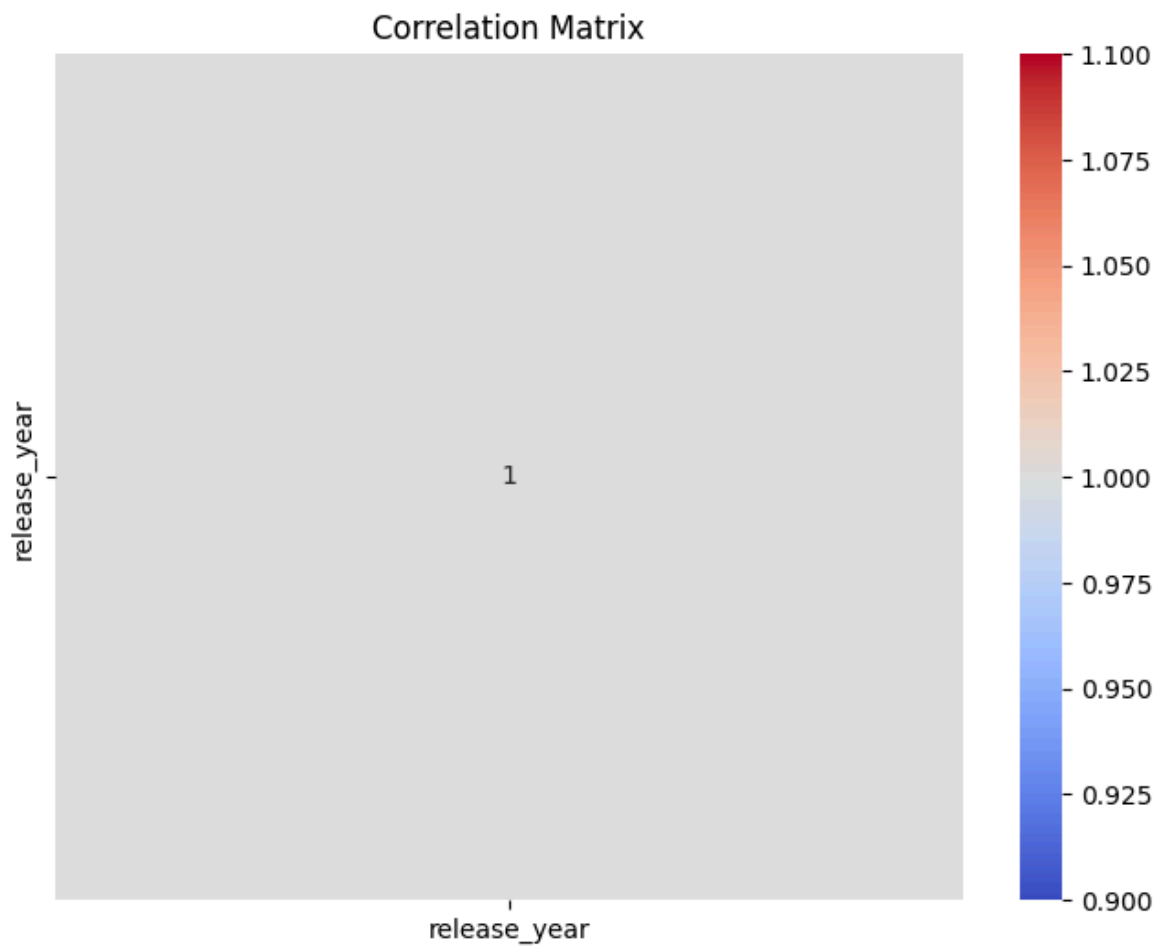
corr = numeric_df.corr()

plt.figure(figsize=(8,6))

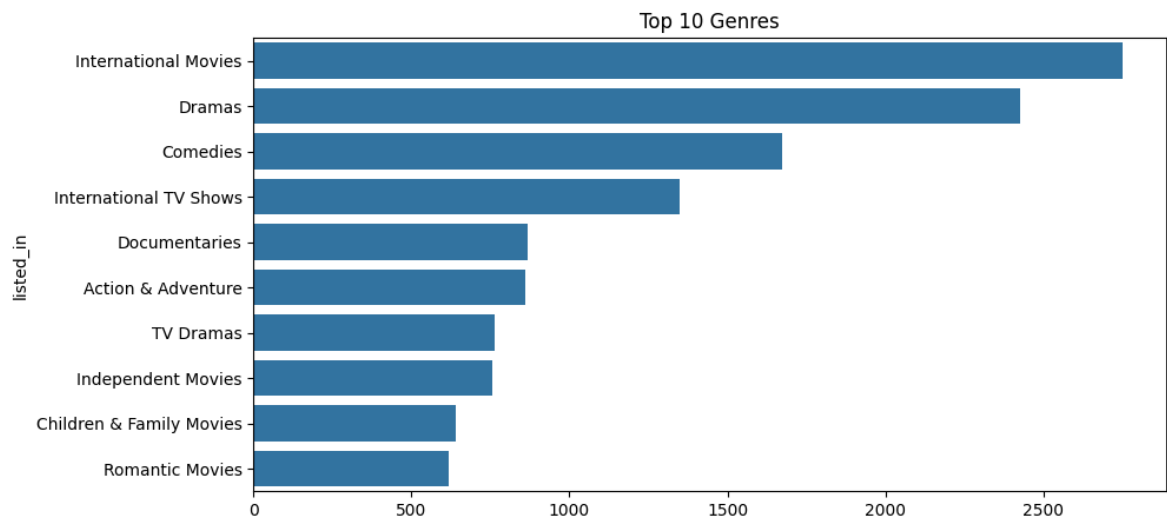
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)
```



```
plt.title("Correlation Matrix")  
plt.show()
```



```
In [13]: genre_counts = df['listed_in'].str.split(', ').explode()  
  
top_genres = genre_counts.value_counts().head(10)  
  
plt.figure(figsize=(10,5))  
  
sns.barplot(  
    x=top_genres.values,  
    y=top_genres.index  
)  
  
plt.title("Top 10 Genres")  
plt.show()
```



In []:

